

1st Set of Problems (Detailed Solution)

The first set of problems comprises a revision of rotations and translations. This revision is complemented through the introduction of the notation that will be used throughout the course. This notation generalizes the composition of rotations and translations, in a manner which is quite useful in the description of robotic manipulators and their configuration.

1.(a) Expressing the components of an *orthonormal frame* (orthogonal frame with unitary vectors) into another is an intuitive approach to the understanding of the geometry of rotations. This is accomplished through the *inner product* between two vectors,

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \vartheta. \quad (1)$$

As illustrated in Figure 1(a), the inner product allows then to express the components of one vector into another. Abbreviating, we call this a *vector projection*.

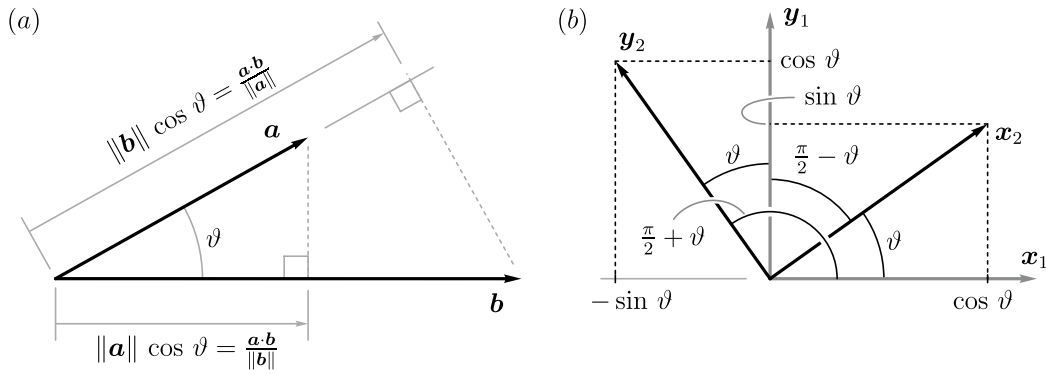


Figure 1: (a) Vector projections. (b) Rotated orthonormal frames.

Returning now to the present problem. If the previous vectors are *unitary* ($\|\mathbf{a}\| = \|\mathbf{b}\| = 1$), then the previous relation reduces to $\mathbf{a} \cdot \mathbf{b} = \cos \vartheta$ and from inspection of Figure 1(b) the vector projections between the orthonormal frames are written as

$$\begin{aligned} \mathbf{x}_2 \cdot \mathbf{x}_1 &= \cos \vartheta, & \mathbf{y}_2 \cdot \mathbf{x}_1 &= \cos \left(\frac{\pi}{2} + \vartheta \right) = -\sin \vartheta, \\ \mathbf{x}_2 \cdot \mathbf{y}_1 &= \cos \left(\frac{\pi}{2} - \vartheta \right) = \sin \vartheta, & \mathbf{y}_2 \cdot \mathbf{y}_1 &= \cos \vartheta. \end{aligned} \quad (2)$$

Note that the trigonometric expressions have been reduced to their first quadrant equivalents, as summarized in Figure 2. Note also that this simplification can be directly inspected from Figure 1(b), without recourse to Equation 1.

Finally, we can express the coordinates of $\mathbf{x}_2$ and $\mathbf{y}_2$ in relation to the first frame, notated as $\mathbf{x}_2^1$ and $\mathbf{y}_2^1$ (respectively). Thus, from inspection of Figure 1(b), the coordinates are

$$\mathbf{x}_2^1 = \begin{pmatrix} \cos \vartheta \\ \sin \vartheta \end{pmatrix}, \quad \mathbf{y}_2^1 = \begin{pmatrix} -\sin \vartheta \\ \cos \vartheta \end{pmatrix}. \quad (3)$$

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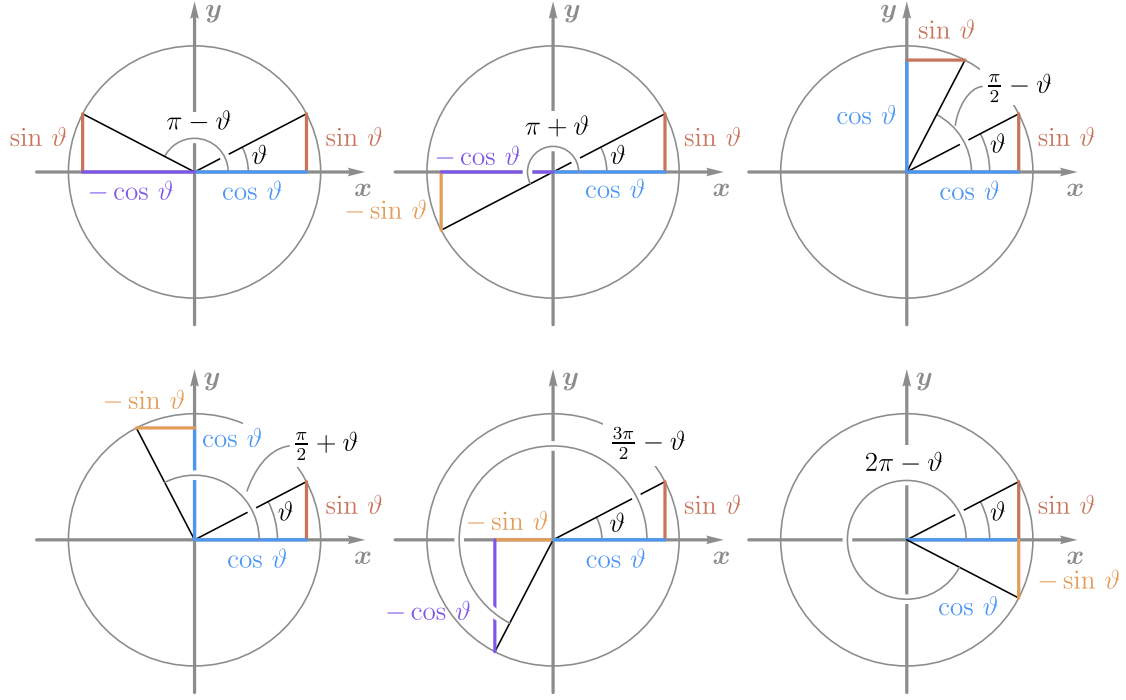


Figure 2: Examples of reductions to the first quadrant.

1.(b) In order to express any vector we must first establish a reference to measure its coordinates. The reference that we commonly use are *right-handed* orthonormal frames in three dimensional space.

To generalize this exercise, and better understand vectors and their representation, consider an arbitrary orthonormal frame α in three dimensional space, $\langle \mathbf{x}_\alpha, \mathbf{y}_\alpha, \mathbf{z}_\alpha \rangle$. Then, the coordinate along the x axis of frame α is notated as r_x^α , the coordinate along the y axis written as r_y^α and the coordinate along the z axis as r_z^α . Thus, the vector can be written as

$$\mathbf{r}^\alpha = \begin{pmatrix} r_x^\alpha \\ r_y^\alpha \\ r_z^\alpha \end{pmatrix}. \quad (4)$$

These coordinates are illustrated in Figure 3(a). It is important to note that in this Figure the superscript α has not been placed in $\mathbf{r}$. Indeed, when drawing the vector it is senseless to attribute a frame specification to a vector. To better understand this, we may first omit the superscript to write the vector as

$$\mathbf{r} = r_x^\alpha \mathbf{x}_\alpha + r_y^\alpha \mathbf{y}_\alpha + r_z^\alpha \mathbf{z}_\alpha. \quad (5)$$

Now, we are free to add another reference to establish the coordinates of the vectors that define frame α . That said, consider another orthonormal frame β , $\langle \mathbf{x}_\beta, \mathbf{y}_\beta, \mathbf{z}_\beta \rangle$, as illustrated in Figure 3(b). Then, if the coordinates of the vectors $\mathbf{x}_\alpha$, $\mathbf{y}_\alpha$ and $\mathbf{z}_\alpha$ are already known with respect to frame β we can also write the coordinates of $\mathbf{r}$ in this frame, through

$$\boxed{\mathbf{r}^\beta = r_x^\alpha \mathbf{x}_\alpha^\beta + r_y^\alpha \mathbf{y}_\alpha^\beta + r_z^\alpha \mathbf{z}_\alpha^\beta.} \quad (6)$$

This result gives us the change in the coordinates of a vector, which is further explored in Problem 1(c).

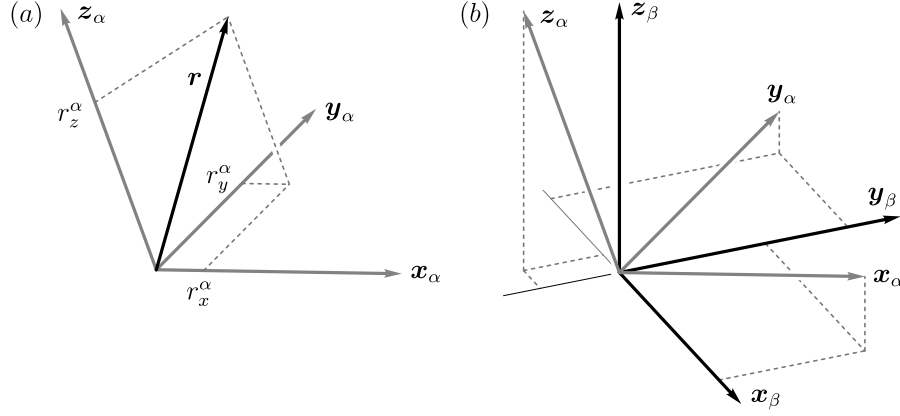


Figure 3: (a) Coordinates of a vector $\mathbf{r}$, expressed in frame α . (b) Coordinates of frame α , expressed in frame β .

Returning now to the present exercise, because the coordinates of frame two in relation to frame one have already been determined in Equation 3, we can apply the previous result to write the coordinates of $\mathbf{r}$ in frame one through

$$\mathbf{r}^1 = r_x^2 \mathbf{x}_2^1 + r_y^2 \mathbf{y}_2^1 = r_x^2 \begin{pmatrix} \cos \vartheta \\ \sin \vartheta \end{pmatrix} + r_y^2 \begin{pmatrix} -\sin \vartheta \\ \cos \vartheta \end{pmatrix}. \quad (7)$$

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As a final note on this exercise, we can ask what are the coordinates of a frame in itself. This question can also be asked through Equations 4 and 6, to write

$$\mathbf{r}^\alpha = \begin{pmatrix} r_x^\alpha \\ r_y^\alpha \\ r_z^\alpha \end{pmatrix} = r_x^\alpha \mathbf{x}_\alpha^\alpha + r_y^\alpha \mathbf{y}_\alpha^\alpha + r_z^\alpha \mathbf{z}_\alpha^\alpha. \quad (8)$$

This allows to conclude that the coordinates of the frame in itself are the *standard basis* coordinates. We commonly notate this basis through the *versors* $\hat{\mathbf{i}} = \mathbf{x}_\alpha^\alpha$, $\hat{\mathbf{j}} = \mathbf{y}_\alpha^\alpha$ and $\hat{\mathbf{k}} = \mathbf{z}_\alpha^\alpha$. Finally, when this versors are gathered into a matrix, their components can be defined as follows,

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (\hat{\mathbf{i}} \quad \hat{\mathbf{j}} \quad \hat{\mathbf{k}}), \quad (9)$$

to say that the standard basis is the identity matrix.

1.(c) In this exercise the *rotation matrix* is finally derived. This is done by writing the result of the previous exercise in matrix form.

Again, we are first going to generalize the exercise. Thus, note that Equation 6 can be written in matrix form through

$$\mathbf{r}^\beta = (\mathbf{x}_\alpha^\beta \quad \mathbf{y}_\alpha^\beta \quad \mathbf{z}_\alpha^\beta) \begin{pmatrix} r_x^\alpha \\ r_y^\alpha \\ r_z^\alpha \end{pmatrix}, \quad (10)$$

where $\begin{pmatrix} x_\alpha^\beta & y_\alpha^\beta & z_\alpha^\beta \end{pmatrix}$ is a three by three matrix. In this matrix, the vectors x_α^β , y_α^β and z_α^β are its column entries, which denote the coordinates of frame α when expressed in frame β . Indeed, this matrix is a rotation matrix, which we can finally define as

$$\mathbf{R}_\alpha^\beta = \begin{pmatrix} x_\alpha^\beta & y_\alpha^\beta & z_\alpha^\beta \end{pmatrix}. \quad (11)$$

Moreover, recalling Equation 4, it is also true that

$$\mathbf{r}^\beta = \mathbf{R}_\alpha^\beta \mathbf{r}^\alpha, \quad (12)$$

which denotes a change in the frame representation of vector $\mathbf{r}$.

The results gathered into Equations 11 and 12 are now directly applied to solve the present exercise. Thus, the rotation matrix that transforms coordinates expressed in frame two into coordinates expressed in frame one is given by

$$\mathbf{R}_2^1 = \begin{pmatrix} x_2^1 & y_2^1 \end{pmatrix} = \begin{pmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{pmatrix}, \quad (13)$$

and the transformation follows

$$\mathbf{r}^1 = \mathbf{R}_2^1 \mathbf{r}^2. \quad (14)$$

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2.(a) In this exercise we derive the elementary rotations illustrated in Figure 4. We do this by first drawing the coordinates of frame two in frame one, as illustrated. Then, Equation 11 is applied to gather the coordinates. For Figure 4(a) the result is

$$\mathbf{R}_1^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{pmatrix}, \quad (15)$$

for Figure 4(b)

$$\mathbf{R}_1^2 = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad (16)$$

and for Figure 4(c)

$$\mathbf{R}_1^2 = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (17)$$

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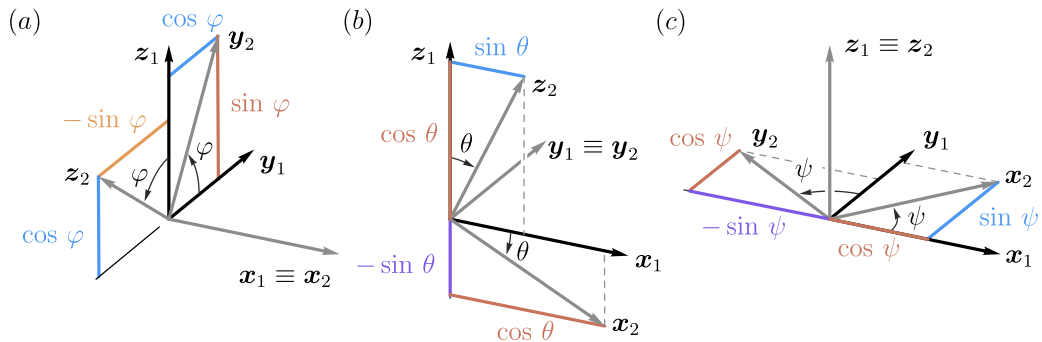


Figure 4: Elementary rotations. (a) Rotation about x by an angle φ . (b) Rotation about y by an angle θ . (c) Rotation about z by an angle ψ .

2.(b) In this exercise we derive another main result relating to rotation matrices, which is the fact that the inverse of a rotation matrix is identical to its transposition. That said, consider the transformation $\mathbf{r}^\beta = \mathbf{R}_\alpha^\beta \mathbf{r}^\alpha$, from which it is desired to express the inverse transformation $\mathbf{r}^\alpha = (\mathbf{R}_\alpha^\beta)^{-1} \mathbf{r}^\beta$. By virtue of the notation, it is true that this transformation can also be written as $\mathbf{r}^\alpha = \mathbf{R}_\beta^\alpha \mathbf{r}^\beta$. Therefore

$$(\mathbf{R}_\alpha^\beta)^{-1} = \mathbf{R}_\beta^\alpha. \quad (18)$$

Now, recall Problem 1(a) to expand Equation 11 as

$$\mathbf{R}_\alpha^\beta = \begin{pmatrix} \mathbf{x}_\alpha \cdot \mathbf{x}_\beta & \mathbf{y}_\alpha \cdot \mathbf{x}_\beta & \mathbf{z}_\alpha \cdot \mathbf{x}_\beta \\ \mathbf{x}_\alpha \cdot \mathbf{y}_\beta & \mathbf{y}_\alpha \cdot \mathbf{y}_\beta & \mathbf{z}_\alpha \cdot \mathbf{y}_\beta \\ \mathbf{x}_\alpha \cdot \mathbf{z}_\beta & \mathbf{y}_\alpha \cdot \mathbf{z}_\beta & \mathbf{z}_\alpha \cdot \mathbf{z}_\beta \end{pmatrix}. \quad (19)$$

Expand also its inverse, to reveal

$$\mathbf{R}_\beta^\alpha = \begin{pmatrix} \mathbf{x}_\beta \cdot \mathbf{x}_\alpha & \mathbf{y}_\beta \cdot \mathbf{x}_\alpha & \mathbf{z}_\beta \cdot \mathbf{x}_\alpha \\ \mathbf{x}_\beta \cdot \mathbf{y}_\alpha & \mathbf{y}_\beta \cdot \mathbf{y}_\alpha & \mathbf{z}_\beta \cdot \mathbf{y}_\alpha \\ \mathbf{x}_\beta \cdot \mathbf{z}_\alpha & \mathbf{y}_\beta \cdot \mathbf{z}_\alpha & \mathbf{z}_\beta \cdot \mathbf{z}_\alpha \end{pmatrix}. \quad (20)$$

Now, because the inner product is commutative, $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$, it must be true that

$$\mathbf{R}_\beta^\alpha = (\mathbf{R}_\alpha^\beta)^\top. \quad (21)$$

Returning to the present exercise, the inverse transformations of the ones obtained in Problem 2(a) are as follows. For Figure 4(a) the result is

$$\mathbf{R}_2^1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{pmatrix}, \quad (22)$$

for Figure 4(b)

$$\mathbf{R}_2^1 = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}, \quad (23)$$

and for Figure 4(c)

$$\mathbf{R}_2^1 = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (24)$$

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3. This exercise motivates the introduction of another important concept, corresponding to the composition of rotations. To see this composition, consider two consecutive transformations. The first being $\mathbf{r}^\gamma = \mathbf{R}_\alpha^\gamma \mathbf{r}^\alpha$ and the second $\mathbf{r}^\beta = \mathbf{R}_\gamma^\beta \mathbf{r}^\gamma$. Then, the composition of these two transformations becomes

$$\mathbf{r}^\beta = \mathbf{R}_\gamma^\beta \mathbf{R}_\alpha^\gamma \mathbf{r}^\alpha. \quad (25)$$

By virtue of the notation, it is true that

$$\boxed{\mathbf{R}_\alpha^\beta = \mathbf{R}_\gamma^\beta \mathbf{R}_\alpha^\gamma} \quad (26)$$

Any number of rotation matrices can be composed in this fashion. For instance, we will see that in a three degrees of freedom robotic manipulator its end-effector orientation is described through the composition $\mathbf{R}_3^o = \mathbf{R}_1^o \mathbf{R}_2^1 \mathbf{R}_3^2$.

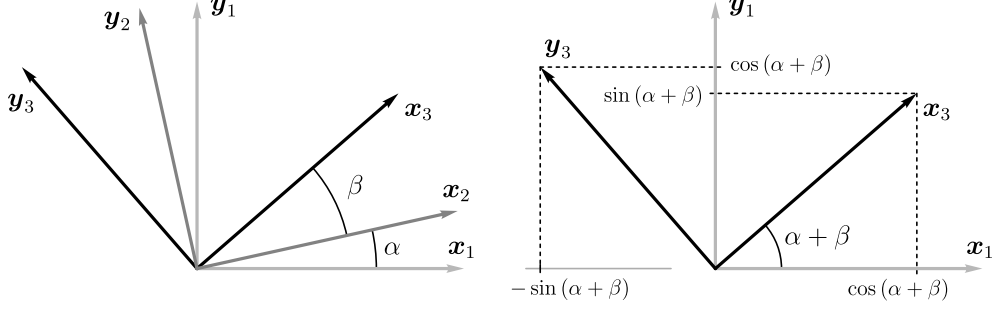


Figure 5: Composition of rotations in the plane.

We can now derive the angle sum identities through the composition of two rotations. The rotations $\mathbf{R}_2^1$ and $\mathbf{R}_3^2$ are written directly from Figure 1(b) and their composition follows¹,

$$\begin{aligned} \mathbf{R}_3^1 &= \mathbf{R}_2^1 \mathbf{R}_3^2 = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} \\ &= \begin{pmatrix} c_\alpha c_\beta - s_\alpha s_\beta & -c_\alpha s_\beta - s_\alpha c_\beta \\ s_\alpha c_\beta + c_\alpha s_\beta & c_\alpha c_\beta - s_\alpha s_\beta \end{pmatrix}. \end{aligned} \quad (27)$$

A similar construction is made in Figure 5, to conclude that

$$\mathbf{R}_3^1 = \begin{pmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{pmatrix}. \quad (28)$$

Identifying Equation 27 with Equation 28 allows then to deduce the following identities,

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta), \quad (29)$$

$$\sin(\alpha + \beta) = \sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta). \quad (30)$$

The remaining identities follow from Figure 2, by noting that $\cos(-\beta) = \cos(\beta)$ and that $\sin(-\beta) = -\sin(\beta)$. Therefore

$$\cos(\alpha - \beta) = \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta), \quad (31)$$

$$\sin(\alpha - \beta) = \sin(\alpha) \cos(\beta) - \cos(\alpha) \sin(\beta). \quad (32)$$

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¹It is usual to use the abbreviations $\cos \alpha = c_\alpha$, $\sin \alpha = s_\alpha$, $\cos(\alpha + \beta) = c_{\alpha\beta}$ and $\sin(\alpha + \beta) = s_{\alpha\beta}$.

4. This exercise motivates the introduction of translations in space and their composition. To describe this composition, consider three points in space, α , β and σ , as illustrated in Figure 6(a). Then, the translation from point α to point σ , notated as $\mathbf{r}_{\alpha,\sigma}$, can be accomplished through²

$$\mathbf{r}_{\alpha,\sigma} = \mathbf{r}_{\alpha,\beta} + \mathbf{r}_{\beta,\sigma}. \quad (33)$$

Translations between arbitrary points can always be summed, as long as their coordinates are expressed in the same frame. Otherwise the coordinates need to be rotated, as expressed in Equation 12. This fact is generally expressed as

$$\mathbf{r}_{\alpha,\sigma}^\gamma = \mathbf{r}_{\alpha,\beta}^\gamma + \mathbf{R}_\beta^\gamma \mathbf{r}_{\beta,\sigma}^\beta. \quad (34)$$

Any number of translations can be composed in this fashion. For instance, in a three degrees of freedom robotic manipulator its end-effector position is described through the composition $\mathbf{p}_3 = \mathbf{r}_{o,3}^o = \mathbf{R}_1^o \mathbf{r}_{o,1}^1 + \mathbf{R}_1^o \mathbf{R}_2^1 \mathbf{r}_{1,2}^2 + \mathbf{R}_1^o \mathbf{R}_2^1 \mathbf{R}_3^2 \mathbf{r}_{2,3}^3$.

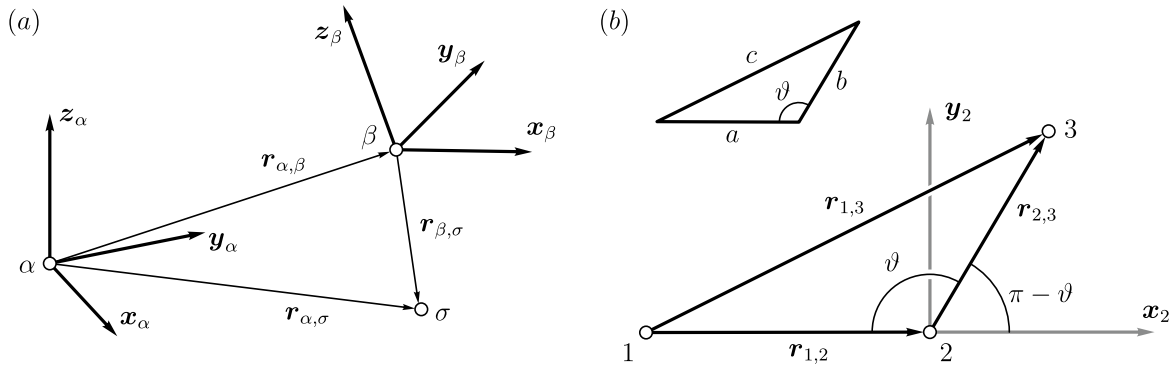


Figure 6: (a) Composition of translations. (b) Law of cosines derivation.

Returning to the present exercise, we can derive the the law of cosines through the composition illustrated in Figure 6(b), where

$$\mathbf{r}_{1,2}^2 = \begin{pmatrix} a \\ 0 \end{pmatrix}, \quad \mathbf{r}_{2,3}^2 = \begin{pmatrix} b \cos(\pi - \vartheta) \\ b \sin(\pi - \vartheta) \end{pmatrix} = \begin{pmatrix} -b \cos \vartheta \\ b \sin \vartheta \end{pmatrix}. \quad (35)$$

Then, from Equation 34, we find that

$$\mathbf{r}_{1,3}^2 = \mathbf{r}_{1,2}^2 + \mathbf{r}_{2,3}^2 = \begin{pmatrix} a - b \cos \vartheta \\ b \sin \vartheta \end{pmatrix}. \quad (36)$$

Now, noting that $\|\mathbf{r}_{1,3}\| = c$, we can compute the norm of the previous vector to find

$$\|\mathbf{r}_{1,3}\|^2 = c^2 = (a - b \cos \vartheta)^2 + b^2 \sin^2 \vartheta = a^2 + b^2 (c_\vartheta^2 + s_\vartheta^2) - 2 a b c_\vartheta. \quad (37)$$

This result can be further simplified by applying Pythagoras' theorem,

$$\cos^2 \vartheta + \sin^2 \vartheta = 1, \quad (38)$$

to conclude that

$$c^2 = a^2 + b^2 - 2 a b c_\vartheta \quad (39)$$

²When the *origin of space* has been established, it is possible to notate the coordinates of a point α as $\mathbf{p}_\alpha$. In which case, the translation from this point to another point β is written as $\mathbf{r}_{\alpha,\beta} = \mathbf{p}_\beta - \mathbf{p}_\alpha$. The origin of space is usually attached to an inertial (unmoving) frame of reference.

List of Formulas (Formulário)

$$\mathbf{R}_\beta^\alpha = \begin{pmatrix} \mathbf{x}_\beta \cdot \mathbf{x}_\alpha & \mathbf{y}_\beta \cdot \mathbf{x}_\alpha & \mathbf{z}_\beta \cdot \mathbf{x}_\alpha \\ \mathbf{x}_\beta \cdot \mathbf{y}_\alpha & \mathbf{y}_\beta \cdot \mathbf{y}_\alpha & \mathbf{z}_\beta \cdot \mathbf{y}_\alpha \\ \mathbf{x}_\beta \cdot \mathbf{z}_\alpha & \mathbf{y}_\beta \cdot \mathbf{z}_\alpha & \mathbf{z}_\beta \cdot \mathbf{z}_\alpha \end{pmatrix}$$

$$\mathbf{R}_\beta^\alpha = \begin{pmatrix} \mathbf{x}_\beta^\alpha & \mathbf{y}_\beta^\alpha & \mathbf{z}_\beta^\alpha \end{pmatrix}$$

$$\mathbf{r}^\alpha = \mathbf{R}_\beta^\alpha \mathbf{r}^\beta$$

$$\mathbf{R}_\beta^\alpha = (\mathbf{R}_\alpha^\beta)^\top$$

$$\mathbf{R}_\beta^\alpha = \mathbf{R}_\gamma^\alpha \mathbf{R}_\beta^\gamma$$

$$\mathbf{R}_\alpha^\alpha = \mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \end{pmatrix}$$

$$\mathbf{r}_{\alpha,\beta} = \mathbf{p}_\beta - \mathbf{p}_\alpha = -\mathbf{r}_{\beta,\alpha}$$

$$\mathbf{r}_{\alpha,\sigma}^\gamma = \mathbf{r}_{\alpha,\beta}^\gamma + \mathbf{R}_\beta^\gamma \mathbf{r}_{\beta,\sigma}^\beta$$